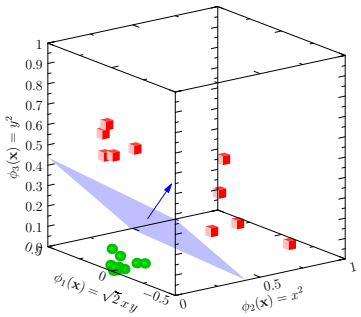
Mathematics of Machine Learning

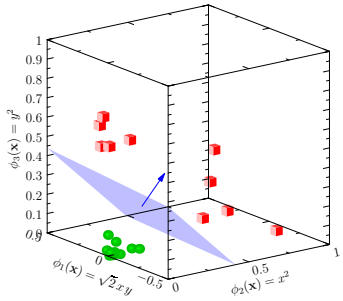
Kernel Trick



The Kernel Trick, SVMs, Regression

Outline

1. **Recap: the Kernel Trick**
2. Mercer's Theorem
3. Constructing Kernels
4. Kernels for Structured Data
5. Beyond Classification



Recap: the Kernel Trick

- SVM kernels are functions of two variables that factorise

$$K(\boldsymbol{x},\boldsymbol{y}) = \langle \boldsymbol{\phi}(\boldsymbol{x}),\boldsymbol{\phi}(\boldsymbol{y})\rangle = \sum_i \phi_i(\boldsymbol{x})\phi_i(\boldsymbol{y})$$

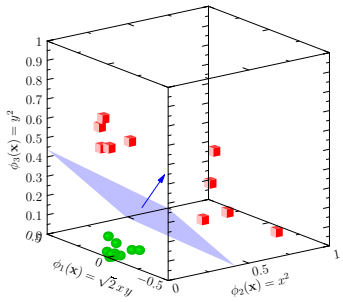
  with real-valued $\phi_i(\boldsymbol{x})$
- Such a K is positive semi-definite and, conversely, any positive semi-definite K factorises (this lecture)
- The SVM dual problem only needs the inner products

$$\max_{\boldsymbol{\alpha}} \sum_{k=1}^m \alpha_k - \frac{1}{2}\sum_{k,l=1}^m \alpha_k\alpha_l y_k y_l K(\boldsymbol{x}_k,\boldsymbol{x}_l)$$

  subject to $\sum_{k=1}^m y_k\alpha_k = 0$ and $0 \le \alpha_k \le C$
- This is the **kernel trick**: we never compute $\boldsymbol{\phi}(\boldsymbol{x})$

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Recap on Eigen Systems

- Recall for a symmetric $(n \times n)$ matrix $\mathbf{M}$ an eigenvector, $\boldsymbol{v}$

$$\mathbf{M}\boldsymbol{v} = \lambda\boldsymbol{v}$$

- There are n independent eigenvectors $\boldsymbol{v}^{(i)}$ with real eigenvalues $\lambda^{(i)}$
- The eigenvectors can be chosen orthonormal: $\boldsymbol{v}^{(i)\mathsf{T}}\boldsymbol{v}^{(j)} = 0$ if $i \neq j$ $(= 1$ if $i = j)$
- Forming a matrix of eigenvectors $\mathbf{V} = (\boldsymbol{v}^{(1)},\boldsymbol{v}^{(2)},\ldots\boldsymbol{v}^{(n)})$ the matrix satisfies

$$\mathbf{V}^\mathsf{T}\mathbf{V} = \mathbf{V}\mathbf{V}^\mathsf{T} = \mathbf{I}$$

- Such matrices are said to be orthogonal

Eigen Decomposition

- From the eigenvalue equation $\mathbf{M}\boldsymbol{v}^{(k)} = \lambda^{(k)}\boldsymbol{v}^{(k)}$

$$\mathbf{M}\mathbf{V} = \mathbf{V}\boldsymbol{\Lambda} \qquad \text{where} \quad \boldsymbol{\Lambda} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

- Multiplying on the right by $\mathbf{V}^\mathsf{T}$ we get

$$\mathbf{M} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^\mathsf{T} = \sum_{k=1}^n \lambda^{(k)}\boldsymbol{v}^{(k)}\boldsymbol{v}^{(k)\mathsf{T}}$$

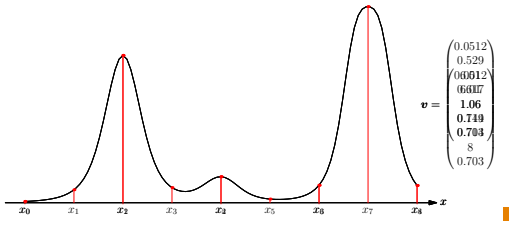
  Or

$$M_{ij} = \sum_{k=1}^n \lambda^{(k)} v_i^{(k)} v_j^{(k)} = \sum_{k=1}^n u_i^{(k)} u_j^{(k)} = \langle \boldsymbol{u}_i,\boldsymbol{u}_j\rangle$$

  $u_i^{(k)} = \sqrt{\lambda^{(k)}}\,v_i^{(k)}$ (real only if $\lambda^{(k)} \ge 0$)

Limit Process

- Consider sampling a function at a set of points

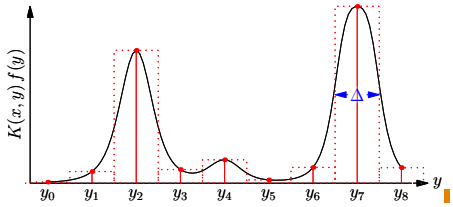


- In the limit where the number of sample points goes to infinity the vector more closely approximates a function
- The kth component of the vector corresponds to $f(x_k)$

Linear Operators

- Recall a linear operator on functions, $\mathcal{T}[f(x)]$, can often be represented by a kernel

$$\mathcal{T}[f(x)] = \int_{y\in\mathcal{I}} K(x,y)f(y)\,\mathrm{d}y \approx \Delta\sum_{j=1}^n K(x,y_j)f(y_j)$$



  This is just a matrix equation with $M_{ij} = \Delta K(x_i,y_j)$

Eigenfunctions

- By analogy for a symmetric function of two variables we can define an *eigenfunction*

$$\int K(\boldsymbol{x},\boldsymbol{y})\psi(\boldsymbol{y})\mathrm{d}\boldsymbol{y} = \lambda\psi(\boldsymbol{x})$$

- In general there will be a denumerable set of eigenfunctions $\psi^{(k)}(\boldsymbol{x})$ where

$$K(\boldsymbol{x},\boldsymbol{y}) = \sum_k \lambda^{(k)}\psi^{(k)}(\boldsymbol{x})\psi^{(k)}(\boldsymbol{y})$$

- For a positive semi-definite kernel (on a bounded domain) this is **Mercer's theorem**

From Eigenfunctions to Features

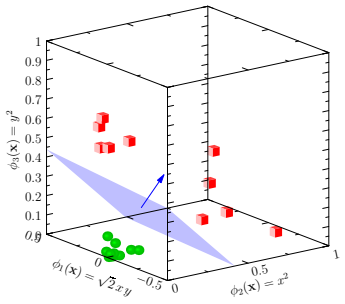
- If we define $\phi^{(k)}(\boldsymbol{x}) = \sqrt{\lambda^{(k)}}\psi^{(k)}(\boldsymbol{x})$ then

$$K(\boldsymbol{x},\boldsymbol{y}) = \sum_k \lambda^{(k)}\psi^{(k)}(\boldsymbol{x})\psi^{(k)}(\boldsymbol{y}) = \sum_k \phi^{(k)}(\boldsymbol{x})\phi^{(k)}(\boldsymbol{y})$$

- This is the definition of an SVM kernel we started with
- Note that for $\phi^{(k)}(\boldsymbol{x})$ to be real $\lambda^{(k)} \geq 0$ for all k
- If $\lambda^{(k)} < 0$ then $\langle\boldsymbol{\phi}(\boldsymbol{x}),\boldsymbol{\phi}(\boldsymbol{x})\rangle = \|\phi(\boldsymbol{x})\|^2$ might be negative and "distance" between points in the extended feature space can be negative!
- If we use a kernel that isn't positive semi-definite then the Hessian of the dual objective need not be negative semi-definite: the objective can be unbounded, with $\boldsymbol{\alpha}$ diverging

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Positive Semi-Definite Kernels

- Kernels (or matrices) that have eigenvalues $\lambda^{(k)} \geq 0$ are called positive semi-definite
- (If the eigenvalues are strictly positive $\lambda^{(k)} > 0$ the kernels or matrices are called positive definite)
- Positive semi-definite kernels can always be decomposed into a sum of real functions

$$K(\boldsymbol{x},\boldsymbol{y}) = \langle\boldsymbol{\phi}(\boldsymbol{x}),\boldsymbol{\phi}(\boldsymbol{y})\rangle$$

Properties of Positive Semi-Definiteness

- Since

$$K(\boldsymbol{x},\boldsymbol{y}) = \big\langle\boldsymbol{\phi}(\boldsymbol{x}),\boldsymbol{\phi}(\boldsymbol{y})\big\rangle$$

- An immediate consequence is that for any function $f(\boldsymbol{x})$

$$\begin{aligned}\int f(\boldsymbol{x})K(\boldsymbol{x},\boldsymbol{y})f(\boldsymbol{y})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{y} &= \int f(\boldsymbol{x})\big\langle\boldsymbol{\phi}(\boldsymbol{x}),\boldsymbol{\phi}(\boldsymbol{y})\big\rangle f(\boldsymbol{y})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{y}\\ &= \left\langle\int f(\boldsymbol{x})\boldsymbol{\phi}(\boldsymbol{x})\mathrm{d}\boldsymbol{x}, \int f(\boldsymbol{y})\boldsymbol{\phi}(\boldsymbol{y})\mathrm{d}\boldsymbol{y}\right\rangle\\ &= \left\|\int f(\boldsymbol{x})\boldsymbol{\phi}(\boldsymbol{x})\mathrm{d}\boldsymbol{x}\right\|^2 \geq 0\end{aligned}$$

Positive Semi-Definiteness

- The following statements are equivalent
  - $K(\boldsymbol{x},\boldsymbol{y})$ is positive semi-definite (written $K(\boldsymbol{x},\boldsymbol{y}) \succeq 0$)
  - The eigenvalues of $K(\boldsymbol{x},\boldsymbol{y})$ are non-negative
  - The kernel can be written

$$K(\boldsymbol{x},\boldsymbol{y}) = \big\langle\boldsymbol{\phi}(\boldsymbol{x}),\boldsymbol{\phi}(\boldsymbol{y})\big\rangle$$

    where the $\phi^{(k)}(\boldsymbol{x})$'s are real functions
  - For any real function $f(\boldsymbol{x})$

$$\int f(\boldsymbol{x})K(\boldsymbol{x},\boldsymbol{y})f(\boldsymbol{y})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{y} \geq 0$$

  - For any points $\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n$ the **Gram matrix** with elements $K_{ij} = K(\boldsymbol{x}_i,\boldsymbol{x}_j)$ is positive semi-definite

Adding Kernels

- We can construct SVM kernels from other kernels
- If $K_1(\boldsymbol{x},\boldsymbol{y})$ and $K_2(\boldsymbol{x},\boldsymbol{y})$ are valid kernels then so is $K_3(\boldsymbol{x},\boldsymbol{y}) = K_1(\boldsymbol{x},\boldsymbol{y}) + K_2(\boldsymbol{x},\boldsymbol{y})$

$$\begin{aligned}Q &= \int f(\boldsymbol{x})K_3(\boldsymbol{x},\boldsymbol{y})f(\boldsymbol{y})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{y}\\ &= \int f(\boldsymbol{x})\big(K_1(\boldsymbol{x},\boldsymbol{y}) + K_2(\boldsymbol{x},\boldsymbol{y})\big)f(\boldsymbol{y})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{y}\\ &= \int f(\boldsymbol{x})K_1(\boldsymbol{x},\boldsymbol{y})f(\boldsymbol{y})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{y} + \int f(\boldsymbol{x})K_2(\boldsymbol{x},\boldsymbol{y})f(\boldsymbol{y})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{y} \geq 0\end{aligned}$$

- If $K(\boldsymbol{x},\boldsymbol{y})$ is a valid kernel so is $cK(\boldsymbol{x},\boldsymbol{y})$ for $c > 0$

Product of Kernels

- If $K_1(\boldsymbol{x},\boldsymbol{y})$ and $K_2(\boldsymbol{x},\boldsymbol{y})$ are valid kernels then so is $K_3(\boldsymbol{x},\boldsymbol{y}) = K_1(\boldsymbol{x},\boldsymbol{y})K_2(\boldsymbol{x},\boldsymbol{y})$
- Writing

$$K_1(\boldsymbol{x},\boldsymbol{y}) = \sum_i \phi_i^{(1)}(\boldsymbol{x})\phi_i^{(1)}(\boldsymbol{y}), \qquad K_2(\boldsymbol{x},\boldsymbol{y}) = \sum_j \phi_j^{(2)}(\boldsymbol{x})\phi_j^{(2)}(\boldsymbol{y})$$

then

$$\begin{aligned}K_3(\boldsymbol{x},\boldsymbol{y}) &= \sum_{i,j} \phi_i^{(1)}(\boldsymbol{x})\phi_i^{(1)}(\boldsymbol{y})\phi_j^{(2)}(\boldsymbol{x})\phi_j^{(2)}(\boldsymbol{y})\\ &= \sum_{i,j} \phi_{ij}^{(3)}(\boldsymbol{x})\phi_{ij}^{(3)}(\boldsymbol{y}) = \left\langle\boldsymbol{\phi}^{(3)}(\boldsymbol{x}),\boldsymbol{\phi}^{(3)}(\boldsymbol{y})\right\rangle\end{aligned}$$

where $\phi_{ij}^{(3)}(\boldsymbol{x}) = \phi_i^{(1)}(\boldsymbol{x})\phi_j^{(2)}(\boldsymbol{x})$

Exponentiating Kernels

- If $K(\boldsymbol{x},\boldsymbol{y})$ is a valid kernel so is $K^n(\boldsymbol{x},\boldsymbol{y})$ (by induction)
  - ⋆ Assume $K(\boldsymbol{x},\boldsymbol{y}) \succeq 0$ this satisfies base case
  - ⋆ If $K^{n-1}(\boldsymbol{x},\boldsymbol{y}) \succeq 0$ then

$$K^n(\boldsymbol{x},\boldsymbol{y}) = K^{n-1}(\boldsymbol{x},\boldsymbol{y})K(\boldsymbol{x},\boldsymbol{y}) \succeq 0$$

- and $\exp(K(\boldsymbol{x},\boldsymbol{y}))$ is also a valid kernel since

$$\mathrm{e}^{K(\boldsymbol{x},\boldsymbol{y})} = \sum_{i=0}^{\infty} \frac{1}{i!} K^i(\boldsymbol{x},\boldsymbol{y}) = 1 + K(\boldsymbol{x},\boldsymbol{y}) + \frac{1}{2}K^2(\boldsymbol{x},\boldsymbol{y}) + \cdots$$

  but each term in the sum is a kernel (the constant 1 has $\phi(\boldsymbol{x}) = 1$)

Polynomial Kernels

- A constant $c \geq 0$ is a kernel: $c = \sqrt{c}\sqrt{c}$, so $\phi(\boldsymbol{x}) = \sqrt{c}$
- $\boldsymbol{x}^\mathsf{T}\boldsymbol{y}$ is a kernel (with $\boldsymbol{\phi}(\boldsymbol{x}) = \boldsymbol{x}$), so the sum $c + \boldsymbol{x}^\mathsf{T}\boldsymbol{y}$ is too
- Hence the **polynomial kernel**

$$K(\boldsymbol{x},\boldsymbol{y}) = \left(c + \boldsymbol{x}^\mathsf{T}\boldsymbol{y}\right)^d$$

  is a kernel for $d = 1, 2, \ldots$
- Its features $\boldsymbol{\phi}(\boldsymbol{x})$ are all the monomials of the x_i of degree up to d (suitably weighted)
- $c = 0$, $d = 2$ is the quadratic kernel from the last lecture

RBF Kernel

- Now $\boldsymbol{x}^\mathsf{T}\boldsymbol{y} = \langle \boldsymbol{x},\boldsymbol{y} \rangle$ is a valid kernel because it is an inner product of functions $\boldsymbol{\phi}(\boldsymbol{x}) = \boldsymbol{x}$
- For $\gamma > 0$ we have $2\gamma\boldsymbol{x}^\mathsf{T}\boldsymbol{y} \succeq 0$
- Thus $\exp(2\gamma\boldsymbol{x}^\mathsf{T}\boldsymbol{y}) \succeq 0$
- If $K(\boldsymbol{x},\boldsymbol{y}) \succeq 0$ then $g(\boldsymbol{x})K(\boldsymbol{x},\boldsymbol{y})g(\boldsymbol{y}) \succeq 0$

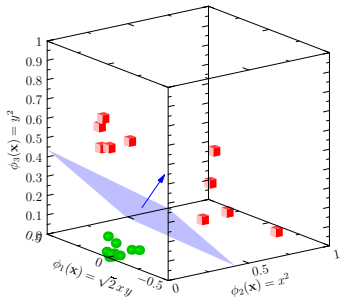
$$\int f(\boldsymbol{x})g(\boldsymbol{x})K(\boldsymbol{x},\boldsymbol{y})g(\boldsymbol{y})f(\boldsymbol{y})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{y} = \int h(\boldsymbol{x})K(\boldsymbol{x},\boldsymbol{y})h(\boldsymbol{y})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{y} \succeq 0$$

  where $f(\boldsymbol{x})g(\boldsymbol{x}) = h(\boldsymbol{x})$

$$\mathrm{e}^{-\gamma\boldsymbol{x}^\mathsf{T}\boldsymbol{x}}\mathrm{e}^{2\gamma\boldsymbol{x}^\mathsf{T}\boldsymbol{y}}\mathrm{e}^{-\gamma\boldsymbol{y}^\mathsf{T}\boldsymbol{y}} = \mathrm{e}^{-\gamma\|\boldsymbol{x}-\boldsymbol{y}\|^2} \succeq 0$$

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Other Kernels

- The success of SVMs has meant that researchers try to increase the area of application
- The condition that an SVM kernel must be positive semi-definite is quite restrictive
- The key to finding new kernels is to use the properties of kernels to build more complicated kernels from simpler ones
- There has been a cottage industry of researchers finding smart kernels for complicated data
- One area where SVMs were very important is document classification, which requires comparing strings

Spectrum Kernel

- A simple way to compare documents is to collect a histogram of all occurrences of substrings of length ℓ
- This is known as an ℓ-spectrum
- The ℓ-spectrum kernel is the inner product of the two histograms

$$s = \texttt{statistics} \quad \mathcal{S}_3(s) = \{\texttt{sta},\texttt{tat},\texttt{ati},\texttt{tis},\texttt{ist},\texttt{sti},\texttt{tic},\texttt{ics}\}$$
$$t = \texttt{computation} \quad \mathcal{S}_3(t) = \{\texttt{com},\texttt{omp},\texttt{mpu},\texttt{put},\texttt{uta},\texttt{tat},\texttt{ati},\texttt{tio},\texttt{ion}\}$$

- $K(s,t) = 1 \times 1 + 1 \times 1 = 2$ ("tat" and "ati" each occur once in both)

All Subsequences Kernel

- A more sophisticated kernel counts all of the common subsequences (not necessarily contiguous) of two documents
- Naively this would take an exponential amount of time to compute
- Using clever dynamic-programming techniques this can be done relatively efficiently
- Gap-weighted versions down-weight subsequences that are spread out over long gaps

Other Kernel Applications

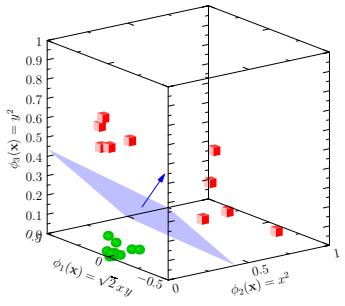
- String kernels for comparing subsequences are used in bioinformatics
- Kernels have been developed for comparing trees (e.g. for computer program evaluation, XML, etc.)
- Kernels have also been developed for comparing graphs (e.g. for comparing chemicals based on their molecular graph)

Fisher Kernels

- In an attempt to build kernels that capture more domain knowledge, kernels are constructed from other learning machines
- An example is the **Fisher kernel**, whose features are the gradients of the log-likelihood of a generative model (e.g. a Hidden Markov Model, HMM) trained on the data
- These tend to have better discriminative power than the underlying model (HMM), and a better feature set than an SVM using a generic kernel

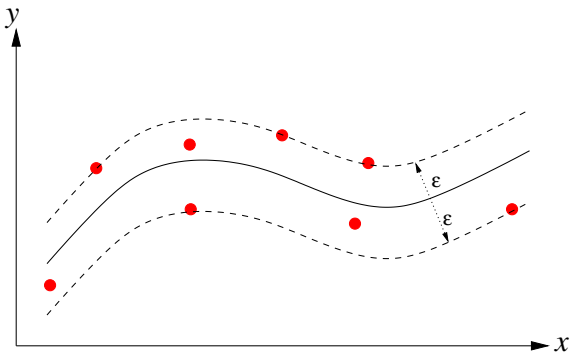
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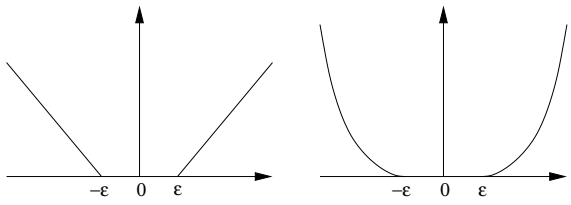
Regression with Margins

- SVMs can be modified to perform regression



Error Functions

- Can introduce slack variables with different errors



- This can be transformed to a quadratic programming problem

Ridge Regression Using Kernels

- We can also solve regression problems without using margins
- Kernel ridge regression minimises

$$\min_{\boldsymbol{w}} \eta\|\boldsymbol{w}\|^2 + \sum_{i=1}^{m}\big(y_i - \boldsymbol{w}^\top\boldsymbol{\phi}(\boldsymbol{x}_i)\big)^2$$

  where $\eta\|\boldsymbol{w}\|^2$ is a regularisation term
- The optimal $\boldsymbol{w}$ lies in the span of the $\boldsymbol{\phi}(\boldsymbol{x}_i)$ (any other component only increases $\|\boldsymbol{w}\|^2$), so $\boldsymbol{w} = \sum_i \alpha_i \boldsymbol{\phi}(\boldsymbol{x}_i)$
- This gives $\boldsymbol{\alpha} = (\mathbf{K} + \eta\mathbf{I})^{-1}\boldsymbol{y}$, with $\mathbf{K}$ the Gram matrix, and predictions $f(\boldsymbol{x}) = \sum_i \alpha_i K(\boldsymbol{x}_i, \boldsymbol{x})$

Kernel Methods

- Kernel methods where we project into an extended feature space are used with other linear algorithms
  - ⋆ Kernel Fisher discriminant analysis (KFDA)
  - ⋆ Kernel principal component analysis (KPCA)
  - ⋆ Kernel canonical correlation analysis (KCCA)
  - ⋆ Gaussian Processes
- These are also extremely powerful machine learning algorithms

Summary

- SVMs require a positive semi-definite kernel function
- These can be built from simpler functions
- There was a cottage industry of people creating new kernels for different applications
- SVMs are just one example of a host of machines that
  - ⋆ use the kernel trick
  - ⋆ often use linear constraints
  - ⋆ tend to be convex optimisation problems